

FINANCIAL RISK ANALYSIS WITH PYTHON

Executive Summary

The project “**Financial Risk Analysis with Python**” focuses on analysing financial transaction data to identify customer behaviour, account activity patterns, potential financial risks, and relationships between transaction activity and account balances.

The analysis was performed using Python and the Pandas, Matplotlib and statistical-analysis capabilities available in Python. The dataset contains **800 financial transaction records**, covering information such as Transaction ID, Customer ID, Account ID, Account Type, Transaction Type, Product, Firm, Region, Manager, Transaction Date, Transaction Amount, Account Balance, Risk Score, Credit Rating and Tenure.

The project was divided into six major analytical tasks:

1. Data Cleaning and Formatting
2. Descriptive Transactional Analysis
3. Customer Profile Building
4. Financial Risk Identification
5. Data Visualisation
6. Hypothesis Testing

The objective was not simply to calculate financial metrics, but to transform transaction-level data into meaningful account-level insights that can support **customer segmentation, risk monitoring, anomaly identification and financial decision-making**.

The analysis identified high- and low-performing accounts, customer activity levels, high-net-inflow accounts, high-frequency low-balance accounts, negative-balance accounts, large withdrawals, high-risk accounts, balance volatility and frequent withdrawal behaviour.

A statistical hypothesis test was also performed to determine whether high-volume transaction accounts have significantly higher average balances than low-volume accounts. The result showed that the difference was **not statistically significant**, indicating that transaction volume alone is not sufficient to explain differences in account balances.

Overall, the project demonstrates how Python-based data analysis can be used to convert raw financial transaction data into structured information for **risk assessment, customer profiling and business decision-making**.

Dataset Overview

The dataset contains **800 transaction records** and **15 original variables**.

The major variables used in the analysis include:

- Transaction ID
- Customer ID
- Account ID
- Account Type
- Transaction Type
- Product
- Firm
- Region
- Manager
- Transaction Date
- Transaction Amount
- Account Balance
- Risk Score

- Credit Rating
- Tenure Months

The dataset contains different transaction categories such as:

- Withdrawal
- Payment
- Deposit
- Transfer

The transaction-type distribution was:

Transaction Type Number of Transactions

Withdrawal	207
Payment	200
Deposit	199
Transfer	194

The account types were:

Account Type Transactions

Loan	218
Credit	206
Savings	204
Current	172

The dataset also contains five regions, with the highest number of transactions coming from the **West region (176)** and the lowest from the **Central region (148)**.

The analysis covers transactions from **2023 to June 2024**.

1. Task 1 – Data Cleaning and Formatting

Objective

The first task was to ensure that the financial dataset was suitable for further analysis.

The data was loaded into a Pandas Data Frame and examined using functions such as:

- head()
- info()
- isnull().sum()
- describe()

The initial dataset contained **800 records and 15 columns**. All columns contained 800 non-null observations, meaning that there were **no missing values in the original dataset**.

The numerical variables were examined using descriptive statistics.

Important overall statistics included:

- Average Transaction Amount: approximately **51,575.67**
- Average Account Balance: approximately **72,345.66**
- Average Risk Score: approximately **0.4734**
- Average Credit Rating: approximately **566.59**
- Average Tenure: approximately **126.58 months**

The minimum observed transaction amount was approximately **–30,721.25**, while the maximum was approximately **130,726.91**.

The minimum account balance was approximately **–37,293.60**, showing that negative account balances were present in the dataset.

The TransactionDate column was converted from text into a proper datetime format using the specified date format:

%d-%m-%Y

This conversion allowed the project to perform monthly and yearly time-based analysis.

Result

The data-cleaning stage produced a structured dataset suitable for transactional, customer-level and risk analysis.

Insight

The data-quality assessment showed that the dataset had complete records with no missing values in the original variables. However, the analysis also revealed negative transaction amounts and negative account balances, which became important indicators during the financial-risk analysis.

2. Task 2 – Descriptive Transactional Analysis

Objective

The objective of Task 2 was to understand how money moves through customer accounts and identify accounts with strong or weak transaction positions.

The analysis focused on:

- Credit and debit transactions
 - Net transaction flow
 - Top-performing accounts
 - Bottom-performing accounts
 - Transaction frequency
 - Transaction gaps
 - Dormant/inactive indicators
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2.1 Credit and Debit Analysis

The project created separate Credit and Debit variables from TransactionAmount.

The calculation used was:

Net Transaction = Credit Amount – Debit Amount

A positive net transaction indicates that credits exceeded debits, while a negative value indicates that debits exceeded credits.

Monthly credit and debit values were then aggregated to identify changes in financial activity over time.

The yearly summary showed:

2023

- Total Credits: approximately **28.52 million**
- Total Debits: approximately **167,805.65**
- Net Transaction Volume: approximately **28.35 million**

2024 up to June

- Total Credits: approximately **13.01 million**
- Total Debits: approximately **101,055.96**
- Net Transaction Volume: approximately **12.91 million**

Because the 2024 data covers only part of the year, direct annual comparison should be interpreted carefully.

Insight

Credit activity was substantially larger than debit activity in the observed dataset. This resulted in positive net transaction volumes throughout the monthly summaries.

This indicates that the dataset is characterised by strong positive credit movement, although individual accounts can still show negative or weak net flows.

2.2 Top Accounts by Net Flow

Account-level NetFlow was calculated by aggregating the difference between credits and debits for every AccountID.

The top-performing account was:

ACC46655 – NetFlow approximately 728,037.40

Other high-net-flow accounts included:

Account	NetFlow
ACC46655	728,037.40
ACC33287	591,591.10
ACC16241	539,612.14
ACC53466	494,957.16
ACC29396	482,720.02
ACC83269	468,734.60
ACC57700	465,047.93
ACC60432	462,957.85
ACC29356	435,573.50
ACC13357	432,527.81

Insight

ACC46655 demonstrated the strongest positive financial movement in the dataset.

A high positive NetFlow can indicate strong inflow activity and potentially valuable customer relationships. However, high NetFlow should be analysed together with transaction frequency, balance and risk score rather than being treated as a standalone indicator of financial strength.

2.3 Bottom Accounts by Net Flow

The bottom accounts were also identified.

The lowest observed NetFlow was:

ACC46953 – approximately -24,811.43

Other accounts with relatively low NetFlow included:

- ACC11062 – approximately 4,014.26
- ACC43771 – approximately 14,033.50
- ACC87602 – approximately 14,944.06
- ACC26956 – approximately 15,603.45

Insight

ACC46953 had more cumulative debit activity than credit activity.

However, a low or negative NetFlow does not automatically mean that an account is financially distressed. It is better interpreted as a **signal for further investigation**.

2.4 Transaction Gaps and Account Activity

Transaction dates were sorted by AccountID and date.

The gap between consecutive transactions was calculated in months.

This allowed the analysis to identify accounts that experienced relatively long periods without recorded transactions.

For example:

- ACC10117 had a maximum transaction gap of **9 months**
- ACC97411 had a maximum transaction gap of **11 months**

The analysis identified prolonged transaction gaps across a significant number of accounts.

Insight

Long transaction gaps can indicate periods of limited account usage. From a business perspective, these accounts may be relevant for:

- Customer re-engagement
- Account monitoring
- Activity analysis
- Further risk investigation

The project uses “Dormant/Inactive” as an **analytical classification**, not as confirmation of an officially dormant bank account.

3. Task 3 – Customer Profile Building

Objective

Task 3 focused on creating meaningful account-level customer profiles using:

- Transaction frequency
- Transaction volume
- Average account balance
- Net inflow

The objective was to understand different customer behaviours and identify specific groups requiring attention.

3.1 Activity-Level Classification

Accounts were classified into three groups based on transaction frequency:

- **Low Activity:** 2 or fewer transactions
- **Medium Activity:** 3–4 transactions
- **High Activity:** More than 4 transactions

The resulting distribution was:

Activity Level Number of Accounts

High	73
Medium	76
Low	45

There were **194 accounts** in total.

Result

Medium-activity accounts formed the largest group, followed closely by high-activity accounts.

Insight

Customer engagement is not uniform.

The large High and Medium activity groups indicate that a substantial portion of accounts are actively used. Low-activity accounts form a smaller but important group that can be monitored for declining engagement or unusual inactivity.

3.2 Customer Segmentation by Balance and Transaction Volume

Two important account-level metrics were calculated:

1. Average Balance
2. Transaction Volume

The median was selected as the segmentation threshold because it is less affected by unusually high or low values.

The calculated thresholds were:

- **Median Average Balance = 72,044.73**
- **Median Transaction Volume = 4 transactions**

Using these thresholds, accounts were divided into four segments:

1. High Balance – High Volume

Customers with relatively high balances and high transaction activity.

These customers can be considered highly engaged and financially significant from an account-management perspective.

2. High Balance – Low Volume

Customers maintaining relatively high balances but conducting fewer transactions.

These accounts may hold substantial funds without frequent transactional activity.

3. Low Balance – High Volume

Customers maintaining relatively lower balances but conducting frequent transactions.

This group is particularly useful for analysing active customers whose balances remain comparatively low.

3.3 Low Balance – Low Volume

Customers with both lower balances and lower transaction activity.

These accounts may require monitoring to understand whether their behaviour is normal or reflects low engagement.

Insight

The segmentation demonstrates that **transaction frequency alone cannot fully describe customer behaviour**.

Combining balance and transaction volume produces a more informative customer profile and can support differentiated customer engagement and risk-monitoring strategies.

3.4 High-Net-Inflow Customer Profile

The project ranked accounts based on total net inflow.

The highest-net-inflow account was:

ACC46655

- Total Net Inflow: **728,037.40**
- Transaction Volume: **14**
- Average Balance: approximately **80,109.06**

Other high-net-inflow accounts included:

- ACC33287 – 591,591.10
- ACC16241 – 539,612.14
- ACC53466 – 494,957.16
- ACC29396 – 482,720.02

Insight

These accounts show strong positive financial movement.

High-net-inflow customers may be important from a relationship-management perspective. However, their transaction patterns should be monitored over time to determine whether the inflows are consistent or temporary.

3.5 High-Frequency, Low-Balance Accounts

The project identified accounts with:

- Transaction volume above the median
- Average balance below the median

Examples included:

Account	Transactions	Average Balance
ACC28292 10		51,228.00
ACC78178 9		69,327.32
ACC88252 8		71,244.26
ACC33287 8		59,331.98
ACC57700 8		60,537.13

Insight

These accounts demonstrate strong transactional activity without maintaining correspondingly high balances.

From a business perspective, these customers may be suitable for targeted engagement and suitable financial-product recommendations.

From a risk perspective, frequent transactions combined with comparatively low balances can also be reviewed for unusual cash-flow patterns.

3.6 Negative or Near-Zero Balance Analysis

The project examined accounts with an average balance of zero or below.

One account was identified:

ACC19178

- Average Balance: approximately **-1,541.18**
- Transaction Volume: **1**
- Segment: **Low Balance – Low Volume**

Insight

A negative average balance may indicate that the account moved below zero during the observed period.

Possible explanations could include overdraft activity, insufficient funds or another account-level condition.

This account therefore represents an account that may require additional monitoring.

The current implementation specifically uses:

AverageBalance ≤ 0

Therefore, small positive balances are not included in the negative/near-zero category unless an explicit near-zero threshold is defined.

4. Task 4 – Financial Risk Identification

Objective

Task 4 focused on identifying potential financial-risk indicators using:

- Large withdrawals
- RiskScore
- Balance volatility
- Statistical withdrawal anomalies
- Frequent withdrawals

- Irregular customer transaction behaviour

The objective was to identify accounts that may require additional monitoring rather than automatically labelling transactions as fraudulent.

4.1 Large Withdrawal Identification

A percentile-based method was used to identify relatively large withdrawals.

Only actual debit transactions were considered.

The **95th percentile** of debit transactions was calculated.

Large Withdrawal Threshold

26,975.89

Therefore, debit transactions equal to or greater than approximately 26,975.89 were classified as relatively large withdrawals.

Two accounts appeared in the resulting account-level summary:

Account	Large Withdrawal Count	Total Large Withdrawals
ACC21719 1		30,721.25
ACC34821 1		27,273.44

Insight

Only two accounts were identified using the 95th-percentile large-withdrawal criterion.

The percentile approach is useful because it identifies unusually large transactions relative to the observed dataset instead of imposing an arbitrary fixed monetary threshold.

4.2 High-Risk Account Identification

Average RiskScore was calculated for each account.

Accounts in the upper 25% of the account-level RiskScore distribution were identified as relatively high-risk accounts.

The highest observed displayed RiskScore was approximately **0.97**, associated with:

ACC49140

Other accounts in the higher-risk group included:

- ACC62809
- ACC43771
- ACC51009
- ACC15671
- ACC87602
- ACC80442
- ACC24880
- ACC39544
- ACC66190

Insight

These accounts fall within the higher-risk portion of the dataset.

However, an important analytical principle is that a high RiskScore **does not independently prove fraud or suspicious behaviour**.

RiskScore should instead be combined with transaction behaviour, balance volatility, withdrawal patterns and other account characteristics.

4.3 Balance Volatility Analysis

Balance volatility was measured using the standard deviation of account balances.

A higher standard deviation indicates that an account's balance changed more substantially across its observed transactions.

Examples included:

- ACC11837 – volatility approximately **60,694.39**
- ACC11285 – volatility approximately **55,922.73**
- ACC11188 – volatility approximately **35,494.66**
- ACC11062 – volatility approximately **3,208.74**

Insight

There was considerable variation in balance volatility between accounts.

Accounts with high balance volatility may warrant additional review, particularly when high volatility occurs together with high RiskScore or unusual withdrawal behaviour.

The volatility calculation produced **16 missing values** because a standard deviation could not be calculated from the available balance observations for those accounts. These missing volatility values should not automatically be interpreted as low risk.

4.4 Withdrawal Anomaly Detection Using IQR

The Interquartile Range (IQR) method was used to detect statistical outliers in withdrawal amounts.

The calculated values were:

- Q1 = **33,317.04**
- Q3 = **73,511.85**
- IQR = **40,194.80**
- IQR Upper Limit = **133,804.05**

Result

Number of anomalous withdrawals = 0

No withdrawal transaction exceeded the IQR-based upper outlier limit.

Insight

According to the standard IQR statistical criterion, the dataset does not contain withdrawal amounts extreme enough to be classified as statistical outliers.

However, zero IQR anomalies does **not** mean that every withdrawal is necessarily normal. It only means that none of the withdrawals was statistically extreme under this particular method.

4.5 Frequent Withdrawal Behaviour

Withdrawal transactions were aggregated by CustomerID.

Customers above the 75th percentile of withdrawal frequency were identified.

The highest withdrawal frequency observed was:

- **CUST1962 – 5 withdrawals**
- **CUST5174 – 5 withdrawals**

Other customers included:

- CUST7173 – 4
- CUST5574 – 4
- CUST5545 – 4
- CUST1121 – 3
- CUST4241 – 3
- CUST2944 – 3
- CUST2694 – 3
- CUST1609 – 3

Insight

Frequent withdrawal activity can be a useful monitoring indicator.

However, frequency alone does not establish suspicious behaviour.

The analysis therefore examined these customers at the transaction level to provide additional context such as:

- Account
- Account Type
- Product
- Region
- Manager
- Transaction Date
- Transaction Amount

Customers such as **CUST1121, CUST1962, CUST2694, CUST2944 and CUST4241** appeared repeatedly in the identified withdrawal activity.

These customers can therefore be considered candidates for further investigation.

4.6 Overall Financial Risk Findings

The financial-risk analysis produced several important findings.

Finding 1 – Large Withdrawals

The 95th-percentile threshold identified withdrawals of approximately **26,975.89 or more** as relatively large.

Two accounts met this criterion.

Finding 2 – High Risk Scores

Accounts in the upper-risk portion of the RiskScore distribution were identified using a percentile-based approach.

Finding 3 – Balance Volatility

Account balances showed substantial differences in volatility, indicating different levels of financial stability across accounts.

Finding 4 – No IQR Withdrawal Outliers

The IQR method identified **zero statistical withdrawal outliers**.

Finding 5 – Frequent Withdrawals

Several customers demonstrated withdrawal frequencies above the 75th percentile, with the highest observed frequency being **5 withdrawals**.

Overall Interpretation

Financial risk should not be assessed using one indicator alone.

A stronger risk-monitoring framework combines:

RiskScore + Withdrawal Behaviour + Balance Volatility + Transaction Frequency + NetFlow + Account Balance

This multi-indicator approach provides a more complete picture of account behaviour.

5. Task 5 – Data Visualisation

Visualisation was used to make the analytical findings easier to understand.

The project created visualisations covering:

1. Transaction Type Distribution
2. Distribution of Transaction Amounts
3. Transaction Amount by Transaction Type
4. Monthly Transaction Trend
5. Monthly Credits vs Debits

6. Transaction Activity by Account Type
7. Transaction Volume by Region
8. Average Transaction Amount by Region
9. Top Customers by Transaction Volume
10. Customer Transaction Frequency vs Average Transaction Amount

These visualisations help identify:

- Which transaction types are most common
- Typical transaction-value ranges
- Extreme transaction values
- Changes in activity over time
- Differences between credits and debits
- Account-type behaviour
- Regional transaction patterns
- Highly active customers
- Relationships between transaction frequency and transaction size

Visualisation Insight

The visual analysis supports the numerical findings by showing that financial behaviour varies across transaction types, accounts, regions and customers.

The combination of charts and numerical analysis makes the project more useful for decision-making because trends and unusual observations can be identified more easily.

6. Task 6 – Hypothesis Testing

Objective

The final task tested whether high-volume transaction accounts have statistically higher average balances than low-volume transaction accounts.

Null Hypothesis (H_0)

High-volume transaction accounts do not have a higher average balance than low-volume transaction accounts.

Alternative Hypothesis (H_1)

High-volume transaction accounts have a statistically higher average balance than low-volume transaction accounts.

The significance level was:

$\alpha = 0.05$

A one-tailed **Welch's independent samples t-test** was selected because it is appropriate when the two groups may have different variances and sample sizes.

6.1 High-Volume and Low-Volume Groups

The 25th and 75th percentiles of transaction volume were used to define clearly separated groups.

The thresholds were:

- Low-volume threshold = **3 transactions**
- High-volume threshold = **5 transactions**

The groups contained:

- **79 low-volume accounts**
- **73 high-volume accounts**

The middle group was excluded from the hypothesis test to create a clearer comparison between relatively low- and high-volume accounts.

6.2 Average Balance Comparison

The descriptive statistics showed:

Group	Average Balance
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Low Volume	71,386.95
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High Volume	72,512.19
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The difference was approximately:

1,125.25

At first glance, high-volume accounts appeared to have a slightly higher average balance. However, statistical testing was required to determine whether this observed difference was meaningful.

6.3 Welch's t-Test Result

The Welch independent samples t-test produced:

- **T-statistic = 0.3391**
- **P-value = 0.367563**
- **Significance Level = 0.05**

The decision rule was:

- If $p\text{-value} < 0.05 \rightarrow \text{Reject } H_0$
- If $p\text{-value} \geq 0.05 \rightarrow \text{Fail to reject } H_0$

Since:

$0.367563 > 0.05$

the null hypothesis was **not rejected**.

Statistical Decision

Fail to reject H_0

Conclusion

There is insufficient statistical evidence to conclude that high-volume transaction accounts have higher average balances than low-volume transaction accounts.

6.4 Effect Size Analysis

Cohen's d was calculated to determine the practical magnitude of the difference.

The result was:

Cohen's d = 0.0539

The effect was classified as:

Negligible effect

This means that even though the high-volume group had a slightly higher average balance, the difference was extremely small from a practical-effect perspective.

Business Insight

Transaction volume alone does not appear to be a strong explanation for differences in average account balances in this dataset.

Therefore, other factors such as account type, customer behaviour, transaction value, RiskScore, tenure and cash-flow patterns should also be considered.

7. Final Project Findings

The complete analysis produced the following major findings:

1. Data Quality

The dataset contained **800 complete transaction records** with no missing values in the original variables.

2. Transaction Behaviour

Withdrawal, payment, deposit and transfer transactions were all represented, with withdrawals being the most frequent transaction type.

3. Net Financial Flow

Most accounts demonstrated positive net transaction flows, while ACC46953 recorded the lowest observed NetFlow at approximately **-24,811.43**.

4. High-Net-Inflow Account

ACC46655 recorded the highest observed NetFlow of approximately **728,037.40** across **14 transactions**.

5. Customer Activity

The 194 accounts were classified into:

- 73 High Activity
- 76 Medium Activity
- 45 Low Activity

6. Customer Segmentation

Using median balance and transaction volume created four meaningful customer segments:

- High Balance – High Volume
- High Balance – Low Volume
- Low Balance – High Volume
- Low Balance – Low Volume

7. High-Frequency Low-Balance Accounts

Several accounts conducted frequent transactions while maintaining below-median balances. These accounts may be useful for both customer-engagement and cash-flow monitoring.

8. Negative Balance

ACC19178 had an average balance of approximately **-1,541.18** and was identified as the negative-average-balance account.

9. Large Withdrawals

The 95th-percentile large-withdrawal threshold was approximately **26,975.89**, identifying two accounts with large withdrawals.

10. High Risk Scores

Accounts in the upper-risk portion of the RiskScore distribution were identified, with the highest displayed score being approximately **0.97**.

11. Balance Volatility

Balance volatility varied significantly across accounts, with some accounts showing substantially larger fluctuations than others.

12. Statistical Withdrawal Anomalies

The IQR method identified **zero withdrawal outliers** under its statistical criterion.

13. Frequent Withdrawal Behaviour

CUST1962 and CUST5174 recorded the highest identified withdrawal frequency of **5 withdrawals**.

14. Hypothesis Testing

The Welch t-test produced a p-value of **0.367563**, which is greater than 0.05.

Therefore, the analysis did not find statistically significant evidence that high-volume accounts have higher average balances.

8. Business Recommendations

Based on the analysis, the following recommendations can be made.

1. Use Multiple Risk Indicators

Financial risk should not be evaluated using RiskScore alone.

RiskScore should be combined with:

- Balance volatility
- Large withdrawals
- Withdrawal frequency
- NetFlow
- Transaction frequency
- Account balance

2. Monitor Negative-Balance Accounts

Accounts with negative average balances should receive additional review to determine whether the condition is caused by overdrafts, insufficient funds or other account-level circumstances.

3. Monitor High-Frequency, Low-Balance Accounts

Frequent transactions with comparatively low balances should be monitored for unusual cash-flow patterns while also being considered for suitable customer-engagement opportunities.

4. Review Large Withdrawals

Transactions exceeding the 95th-percentile threshold should be reviewed together with account history and customer behaviour rather than automatically being considered suspicious.

5. Re-engage Low-Activity Customers

Low-activity accounts can be monitored for declining engagement and considered for suitable customer-retention strategies.

6. Investigate Repeated Withdrawal Behaviour

Customers with frequent withdrawals can be subjected to additional transaction-level review, particularly when multiple risk indicators occur together.

7. Avoid Overinterpreting Statistical Results

The hypothesis test demonstrates that higher transaction volume does not necessarily imply higher account balances.

Therefore, financial decisions should be based on a broader set of account characteristics.

9. Analytical Limitations

The project also has several important limitations.

First, the dataset represents a specific observed period, so the findings should not automatically be generalised to all customers or future periods.

Second, the RiskScore is treated as an existing variable in the dataset. The project analyses the score but does not construct or validate the underlying scoring model.

Third, statistical outlier detection depends on the selected methodology. The fact that the IQR method identified zero withdrawal outliers does not prove that every withdrawal is normal.

Fourth, the “Dormant/Inactive” classification is an analytical classification based on observed transaction gaps and should not be interpreted as an official banking-status designation.

Fifth, the negative/near-zero balance analysis currently uses an average balance of zero or below. Small positive balances are therefore not included as near-zero unless a separate threshold is defined. Finally, statistical significance and business significance are different concepts. The hypothesis test found a negligible effect, showing that the observed difference between high- and low-volume accounts is not practically important in this dataset.

10. Overall Conclusion

The **Financial Risk Analysis with Python** project successfully demonstrates the application of Python-based data analytics to a financial transaction dataset.

The project began with data cleaning and formatting and progressed through descriptive analysis, customer profiling, risk identification, visualisation and statistical hypothesis testing.

The analysis revealed significant variation in customer activity, account balances, transaction frequency and financial behaviour. High-net-inflow accounts such as **ACC46655** were identified, while low-net-flow and negative-balance accounts were also detected.

Customer profiling demonstrated that transaction frequency alone is not sufficient to understand financial behaviour. Combining average balance with transaction volume created four useful customer segments and highlighted high-frequency, low-balance customers.

The financial-risk analysis identified large withdrawals, relatively high-risk accounts, balance volatility and frequent withdrawal behaviour. Importantly, these indicators were treated as **signals requiring further investigation rather than definitive evidence of fraud or suspicious activity**.

The final hypothesis test provided an important statistical conclusion. Although high-volume accounts had a slightly higher average balance than low-volume accounts, the difference was not statistically significant. The Welch's t-test produced a p-value of **0.367563**, while Cohen's d was only **0.0539**, indicating a negligible effect.

Therefore, the overall conclusion is that **transaction volume alone does not sufficiently explain differences in account balances or financial risk**.

A more effective financial-risk framework should combine multiple dimensions including transaction behaviour, account balance, RiskScore, withdrawal activity, balance volatility, customer activity and net financial flow.

Overall, the project demonstrates how Python can transform raw financial transaction data into meaningful analytical insights that can support **customer segmentation, risk monitoring, anomaly detection and evidence-based financial decision-making**.